

FCAI ANU Materials – Semi Seniors – Chapter 2 – Visual Programming

GUI Application Quiz Questions

1. What is the purpose of the TextBox control in a GUI application?

- A. To display static text
- B. To accept keyboard input from the user
- C. To perform calculations
- D. To close the application

Answer: B

2. Where is the TextBox tool located in the Toolbox?

- A. In the Data group
- B. In the Common Controls group
- C. In the Containers group
- D. In the Menus & Toolbars group

Answer: B

3. How is a TextBox control created on the form?

- A. By single-clicking the tool
- B. By double-clicking the tool
- C. By dragging the tool
- D. By right-clicking the tool

Answer: B

4. In which property is the text typed into a TextBox control stored?

- A. Name property
- B. Value property
- C. Text property
- D. Input property

Answer: C

5. What data type is always returned when retrieving the contents of a TextBox's Text property?

- A. Integer
- B. Double
- C. String
- D. Boolean

Answer: C

6. In Example 1, what happens when the user clicks the readInputButton?

- A. The application closes
- B. The user's name is displayed in the outputLabel
- C. The TextBox is cleared
- D. A message box appears

Answer: B

7. What statement is used in Example 1 to display the user's name in the outputLabel?

- A. `outputLabel.Text = "Hello";`
- B. `outputLabel.Text = nameTextBox.Text;`
- C. `nameTextBox.Text = outputLabel.Text;`
- D. `MessageBox.Show(nameTextBox.Text);`

Answer: B

8. How do you clear the contents of a TextBox control?

- A. Set Text property to null
- B. Set Text property to 0
- C. Set Text property to ""
- D. Set Text property to "clear"

Answer: C

9. What is the statement to clear the nameTextBox control?

- A. nameTextBox.Text = null;
- B. nameTextBox.Text = "";
- C. nameTextBox.Clear();
- D. nameTextBox.Text = 0;

Answer: B

10. In Example 1, what happens when the user clicks the exitButton?

- A. The name is displayed
- B. The application is closed
- C. The TextBox is cleared
- D. A calculation is performed

Answer: B

11. What operation appends one string to the end of another?

- A. Subtraction
- B. Concatenation
- C. Division
- D. Multiplication

Answer: B

12. Which operator is used in C# to concatenate strings?

- A. -
- B. *
- C. /
- D. +

Answer: D

13. What is the result of the following code: string message = "Hello " + "world";

- A. "Helloworld"
- B. "Hello world"
- C. "Hello+world"
- D. Error

Answer: B

14. In the string concatenation example, what is displayed in the message box?

- A. Hello
- B. world
- C. Hello world
- D. Hello + world

Answer: C

15. What does the + operator do when both operands are strings?

- A. Adds them numerically
- B. Concatenates them
- C. Subtracts them
- D. Multiplies them

Answer: B

16. In Example 2, how many TextBoxes are there for entering names?

- A. 1
- B. 2
- C. 3
- D. 4

Answer: B

17. What is concatenated in Example 2 to form the full name?

- A. First name and last name without space
- B. First name, space, last name

- C. Last name, space, first name
- D. Only first name

Answer: B

18. What variable is used in Example 2 to store the concatenated full name?

- A. name
- B. fullName
- C. firstName
- D. lastName

Answer: B

19. In Example 2, if first name is Ahmed and last name is Atif, what is displayed?

- A. AhmedAtif
- B. Atif Ahmed
- C. Ahmed Atif
- D. AhmedAtif

Answer: C

20. What button in Example 2 displays the full name?

- A. exitButton
- B. showNameButton
- C. readInputButton
- D. clearButton

Answer: B

21. What are variables declared inside a method called?

- A. Global variables
- B. Local variables
- C. Constant variables
- D. Field variables

Answer: B

22. Can a local variable be accessed from another method?

- A. Yes
- B. No
- C. Only if public
- D. Only if static

Answer: B

23. What term describes the part of a program where a variable can be accessed?

- A. Lifetime
- B. Scope
- C. Type
- D. Value

Answer: B

24. Where does a local variable's scope begin?

- A. At the end of the method
- B. At the beginning of the program
- C. At the variable's declaration
- D. At the class level

Answer: C

25. Where does a local variable's scope end?

- A. At the variable's declaration
- B. At the end of the method where declared
- C. At the end of the program
- D. At the beginning of the method

Answer: B

26. Can multiple variables with the same name be declared in the same method?

- A. Yes
- B. No
- C. Only if different types
- D. Only if constants

Answer: B

27. Can variables with the same name be declared in different methods?

- A. Yes
- B. No
- C. Only if local
- D. Only if fields

Answer: A

28. What happens to a variable's value when a different value is assigned?

- A. It remains the same
- B. It is appended
- C. It is replaced
- D. It causes an error

Answer: C

29. In the duplicate variable names example, what is the final value displayed?

- A. Large Medium-Roast Coffee
- B. Chocolate Truffle
- C. Medium-Roast Coffee
- D. Chocolate

Answer: B

30. Why does the code `employeeID = 125;` cause an error for a string variable?

- A. 125 is too large
- B. It is a nonstring value
- C. String can't hold numbers
- D. Missing quotes

Answer: B

31. How can you store 125 in a string variable?

- A. `employeeID = 125;`
- B. `employeeID = "125";`
- C. `employeeID = int.Parse(125);`
- D. `employeeID = 125.ToString();`

Answer: B

32. In Example 3, what information is entered by the user?

- A. Name and address
- B. Birthdate details
- C. Test scores
- D. Miles and gallons

Answer: B

33. How is the date string formed in Example 3?

- A. Concatenating with no separators
- B. Concatenating day of week, month, day, year with commas and spaces
- C. Using addition
- D. Parsing numbers

Answer: B

34. What button clears the TextBoxes in Example 3?

- A. Show Date
- B. Clear
- C. Exit
- D. Calculate

Answer: B

35. If day of week is Friday, month June, day 1, year 1990, what is displayed?

- A. FridayJune11990
- B. Friday, June 1, 1990
- C. 1 June Friday 1990
- D. 1990, June 1, Friday

Answer: B

36. How do you declare multiple string variables in one statement?

- A. string lastName; firstName; middleName;
- B. string lastName, firstName, middleName;
- C. string lastName firstName middleName;
- D. string (lastName, firstName, middleName);

Answer: B

37. Can you initialize multiple variables in one declaration statement?

- A. No
- B. Yes, like string a = "x", b = "y";
- C. Only if same value
- D. Only for ints

Answer: B

38. Can you assign a double value to an int variable without casting?

- A. Yes
- B. No

- C. Only if small
- D. Only if positive

Answer: B

39. Can you assign an int to a double variable?

- A. No
- B. Yes, implicitly
- C. Only with cast
- D. Only if zero

Answer: B

40. Why can't you assign a decimal to a double variable?

- A. Decimal has greater precision
- B. Double is larger
- C. Types are incompatible
- D. Decimal is integer

Answer: A

41. What is used to explicitly convert numeric types even with data loss?

- A. Parse method
- B. ToString method
- C. Cast operator
- D. + operator

Answer: C

42. What is the syntax for a cast operator?

- A. (type)value
- B. type(value)
- C. value as type
- D. Convert.ToType(value)

Answer: A

43. What is the result of `int whole = (int)3.9;`

- A. 3.9
- B. 4
- C. 3
- D. Error

Answer: C

44. In mixed expressions, what happens when `int` and `double` are operated?

- A. Error
- B. `int` converted to `double`, result `double`
- C. `double` converted to `int`, result `int`
- D. Both stay same

Answer: B

45. Are `double` and decimal mixed expressions allowed without cast?

- A. Yes
- B. No
- C. Only addition
- D. Only multiplication

Answer: B

46. How to fix `double * decimal`?

- A. Cast decimal to `double`
- B. Cast one to the other
- C. Use `Parse`
- D. Impossible

Answer: B

47. Data entered in TextBox is stored as what type in Text property?

- A. The type entered
- B. Always string
- C. Int if number
- D. Double if decimal

Answer: B

48. To store TextBox input as number, what must be done?

- A. Direct assignment
- B. Use cast
- C. Use Parse method
- D. Use ToString

Answer: C

49. Which method converts string to int?

- A. int.ToInt
- B. int.Parse
- C. Parse.int
- D. String.ToInt

Answer: B

50. To display a numeric variable in a Label, what must be done?

- A. Direct assignment
- B. Use Parse
- C. Convert to string with ToString
- D. Use cast

Answer: C

51. What does `variable.ToString()` do?

- A. Converts to int
- B. Converts to string
- C. Formats number
- D. Parses string

Answer: B

52. If one operand is string and other number in `+` operation, what happens?

- A. Error
- B. Number to string, concatenate
- C. String to number, add
- D. Depends on order

Answer: B

53. In Example 4, what is calculated?

- A. Test average
- B. Sale price
- C. MPG
- D. Birth date

Answer: C

54. What types are used for miles and gallons in Example 4?

- A. Int
- B. String
- C. Double
- D. Decimal

Answer: C

55. How is MPG displayed in Example 4?

- A. As double
- B. With ToString()
- C. With Parse
- D. Directly

Answer: B

56. What format string for currency?

- A. "n"
- B. "c"
- C. "p"
- D. "e"

Answer: B

57. What does "n" format do?

- A. Currency
- B. Number with commas
- C. Percent
- D. Exponential

Answer: B

58. How many decimal places by default in currency format?

- A. 0
- B. 1
- C. 2
- D. 3

Answer: C

59. In Example 5, what is calculated?

- A. MPG
- B. Average score

- C. Sale price after discount
- D. Full name

Answer: C

60. How is discount percentage converted to decimal?

- A. Multiply by 100
- B. Divide by 100
- C. Parse
- D. Cast

Answer: B

61. How is sale price displayed in Example 5?

- A. ToString("n")
- B. ToString("c")
- C. ToString("p")
- D. Direct

Answer: B

62. What is an exception?

- A. Normal operation
- B. Unexpected error
- C. User input
- D. Variable declaration

Answer: B

63. What statement handles exceptions?

- A. if-else
- B. try-catch
- C. for loop
- D. while

Answer: B

64. In try-catch, where is potentially error-prone code placed?

- A. In catch
- B. In try
- C. Outside
- D. In finally

Answer: B

65. What happens if exception in try block?

- A. Program crashes
- B. Execution jumps to catch
- C. Continues normally
- D. Skips try

Answer: B

66. In Example 6, what is handled?

- A. Division by zero
- B. Invalid input for Parse
- C. File not found
- D. Overflow

Answer: B

67. What are fields?

- A. Local variables
- B. Variables declared in class but not in methods
- C. Constants
- D. Parameters

Answer: B

68. What is the scope of a field?

- A. Method
- B. Entire class
- C. Block
- D. Program

Answer: B

69. What access modifier for fields by default?

- A. Public
- B. Private
- C. Protected
- D. Internal

Answer: B

70. In Example 7, where is the name variable declared?

- A. In method
- B. As local
- C. As field
- D. As constant

Answer: C

71. What is a constant field?

- A. Can be changed
- B. Cannot be changed after initialization
- C. Local constant
- D. Global variable

Answer: B

72. How to declare constant field?

- A. `const type name = value;`
- B. `constant type name = value;`
- C. `type name = const value;`
- D. `type const name = value;`

Answer: A

73. In Example 8, what is used for the banknote values?

- A. Local variables
- B. Fields
- C. Constants
- D. Parameters

Answer: C

74. What does `Math.Abs` do?

- A. Square root
- B. Absolute value
- C. Power
- D. Max

Answer: B

75. What does `Math.Pow(x, y)` return?

- A. $x + y$
- B. $x ^ y$
- C. $y ^ x$
- D. $x * y$

Answer: B

76. When does a control have focus?

- A. When clicked
- B. Receives keyboard input
- C. When hovered
- D. Always

Answer: B

77. How to tell which control has focus?

- A. Color change
- B. Blinking cursor in TextBox or dotted line on button
- C. Size change
- D. Invisible

Answer: B

78. What changes focus?

- A. Enter key
- B. Tab key
- C. Space key
- D. Alt key

Answer: B

79. What property controls tab order?

- A. FocusIndex
- B. TabOrder
- C. TabIndex
- D. OrderTab

Answer: C

80. Lowest TabIndex is?

- A. Last in order
- B. First in order

- C. Irrelevant
- D. Negative

Answer: B

81. How to set tab order visually?

- A. View > Tab Order
- B. Edit > Tab
- C. Tools > Order
- D. Format > Tab

Answer: A

82. What method moves focus?

- A. SetFocus()
- B. Focus()
- C. GetFocus()
- D. MoveFocus()

Answer: B

83. What is access key?

- A. Key alone
- B. Alt + key
- C. Ctrl + key
- D. Shift + key

Answer: B

84. How to assign access key to button?

- A. & in Text before letter
- B. * before letter
- C. # before letter
- D. @ before letter

Answer: A

85. To display & on button?

- A. &
- B. &&
- C. &
- D. * &

Answer: B

86. What property changes background color?

- A. ForeColor
- B. BackColor
- C. ColorBack
- D. Background

Answer: B

87. What property changes text color?

- A. TextColor
- B. ForeColor
- C. ColorFore
- D. FontColor

Answer: B

88. How to set color in code?

- A. `Control.BackColor = Color.Red;`
- B. `Control.BackColor("Red");`
- C. `Color.Set(Control, Red);`
- D. `Control.Color = Red;`

Answer: A

89. Property for form background image?

- A. ImageBack
- B. BackgroundImage
- C. BackImage
- D. FormImage

Answer: B

90. Default BackgroundImageLayout?

- A. None
- B. Tile
- C. Center
- D. Stretch

Answer: B

91. Which layout resizes without stretching?

- A. Stretch
- B. Tile
- C. Zoom
- D. None

Answer: C

92. What is GroupBox?

- A. Text display
- B. Container with title
- C. Button group
- D. Image holder

Answer: B

93. Does GroupBox have Text property?

- A. No
- B. Yes, for title
- C. For content
- D. For border

Answer: B

94. How tab order in GroupBox?

- A. Independent
- B. Relative to GroupBox TabIndex
- C. Global
- D. None

Answer: B

95. Difference between Panel and GroupBox?

- A. Panel has title
- B. Panel no title, has BorderStyle
- C. Same
- D. Panel not container

Answer: B

96. Default BorderStyle for Panel?

- A. FixedSingle
- B. Fixed3D
- C. None
- D. Tile

Answer: C

97. In exception handling, what is created when exception thrown?

- A. Variable
- B. Exception object
- C. Method
- D. Class

Answer: B

98. How to name exception object?

- A. catch(ex)
- B. catch(Exception ex)
- C. try(ex)
- D. exception ex

Answer: B

99. In formatting, what is "p" for?

- A. Percent
- B. Currency
- C. Number
- D. Fixed

Answer: A

100. What does Math.Min return?

- A. Maximum
- B. Minimum
- C. Average
- D. Sum

Answer: B

Wish you all the best!

Fathy.